2023 SSMO Speed Round Solutions

SMO Team

SR 1: Let $S_1 = \{2, 0, 3\}$ and $S_2 = \{2, 20, 202, 2023\}$. Find the last digit of

$$\sum_{a \in S_1, b \in S_2} a^b.$$

Answer: $\boxed{8}$

Solution: Since $0^b = 0$ for every positive integer b, we want the last digit of

$$2^2 + 2^{20} + 2^{202} + 2^{2023} + 3^2 + 3^{20} + 3^{202} + 3^{2023}.$$

Powers of 2 (and of 3) are periodic mod 10 with period 4, so the four exponents above are congruent mod 4 to $2, 0, 2, 3$ respectively. Hence the sum has the same last digit as

$$2^2 + 2^4 + 2^2 + 2^3 + 3^2 + 3^4 + 3^2 + 3^3 = 4 + 6 + 4 + 8 + 9 + 1 + 9 + 7 = 48,$$

so the last digit is $\boxed{8}$.

SR 2: Points A, B, C are chosen independently and uniformly at random inside the square with vertices $(-1, -1)$, $(-1, 1)$, $(1, -1)$, and $(1, 1)$. The probability that the centroid of triangle ABC lies in the first quadrant can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{5}$

Solution: Let $A = (a_1, a_2)$, $B = (b_1, b_2)$, and $C = (c_1, c_2)$. Note that all coordinates are independently and uniformly distributed on $[-1, 1]$. We want the probability that $a_1+b_1+c_1 \geq 0$ and $a_2+b_2+c_2 \geq 0$ both hold. These are independent events. Since (a_1, b_1, c_1) and $(-a_1, -b_1, -c_1)$ are equally likely, by symmetry, $a_1+b_1+c_1 \geq 0$ and $a_1+b_1+c_1 \leq 0$ are equally likely. The case $a_1+b_1+c_1 = 0$ occurs with probability 0, so $P(a_1+b_1+c_1 \geq 0) = \frac{1}{2}$. The same holds for $a_2 + b_2 + c_2$. Therefore, the desired probability is

$$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \implies 1 + 4 = \boxed{5}.$$

SR 3: Pigs like to eat carrots. Suppose a pig chooses 6 letters one at a time, independently and uniformly at random from the set $\{c, a, r, o, t\}$, to form a "word". If the 6 letters do not spell "carrot", the pig gets frustrated and tries to spell it again (by rechoosing all 6 letters). What is the expected number of tries it takes for the pig to spell "carrot"?

Answer: $\boxed{15625}$

Solution: Note that each of the 5^6 possible words is equally likely, so each try spells "carrot" with probability $\frac{1}{5^6}$. Let x be the expected number of tries. We have

$$x = 1 + \frac{5^6 - 1}{5^6} \cdot x \implies x = 5^6 = \boxed{15625}.$$

SR 4: Let $F_1 = F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for all $n \geq 3$ be the Fibonacci numbers. If distinct positive integers $a_1, a_2, \ldots, a_m$ satisfy $F_{a_1} + F_{a_2} + \cdots + F_{a_m} = 2023$, find the minimum possible value of $a_1 + a_2 + \cdots + a_m$.

Answer: $\boxed{50}$

Solution: Suppose that $a_1 < a_2 < \cdots < a_m$ are chosen such that $a_1 + a_2 + \cdots + a_m$ is minimal and

$$F_{a_1} + F_{a_2} + \cdots + F_{a_m} = 2023.$$

Then, we may assume there are no consecutive indices, meaning $a_i + 1 \neq a_{i+1}$ for all i, since replacing $F_a + F_{a+1}$ with F_{a+2} does not increase the sum of the indices. We generalize 2023 to any $k > 1$. Let a be maximal such that $F_a \leq k$, and we claim that $a_m = a$. If k is a Fibonacci number, the result is immediate. It is easy to show by induction that

$$F_{2n+1} = F_{2n} + F_{2n-2} + \cdots + F_2 + 1 \quad \text{and} \quad F_{2n+2} = F_{2n+1} + F_{2n-1} + \cdots + F_1.$$

Thus, if $a_m \neq a$, then

$$F_{a_1} + F_{a_2} + \cdots + F_{a_m} \leq F_a < k,$$

a contradiction. Therefore, a_m must be maximal, and we may reduce greedily to get

$$2023 = F_{17} + F_{14} + F_9 + F_7 + F_3,$$

giving an answer of

$$17 + 14 + 9 + 7 + 3 = \boxed{50}.$$

SR 5: In parallelogram $ABCD$ with $AB = 6$ and $BC = 8$, a point P inside the parallelogram is chosen such that $\angle APD + \angle BPC = 180^\circ$. Find the sum of the maximum and minimum values of $(PA)(PC) + (PB)(PD)$.

Answer: $\boxed{96}$

Solution: A translation that takes BC to AD takes P to P'. Thus, $\angle AP'D + \angle APD = \angle BPC + \angle APD = 180^\circ$, meaning $PAP'D$ is cyclic. From Ptolemy's Theorem,

$$(AD)(PP') = (PA)(P'D) + (PD)(P'A).$$

Since $P'D = PC$, $P'A = PB$, and $PP' = AB$, we have

$$(PA)(PC) + (PB)(PD) = (AD)(AB) = 8 \cdot 6 = 48$$

for every such P. So, the maximum and minimum are both 48, giving an answer of $48+48 = \boxed{96}$.

SR 6: Find the smallest odd prime that does not divide $2^{75!} - 1$.

Answer: $\boxed{167}$

Solution: Let this odd prime be p. Note that $2^{75!}-1$ is divisible by p if and only if

$$2^{75!} \equiv 1 \pmod{p},$$

or equivalently $\text{ord}_p(2) \mid 75!$. Since $\text{ord}_p(2) \mid p-1$, this holds whenever $p-1 \mid 75!$, which is the case for any $p < 167$ such that every prime factor of $p-1$ is at most 75. So, $p-1$ must have a prime factor $q > 75$. Since $p-1$ is even, we have $p-1 \geq 2q$, and the smallest prime of this form is $p = 2 \cdot 83 + 1 = 167$ (as $2 \cdot 79 + 1 = 159 = 3 \cdot 53$). Now, $\text{ord}_{167}(2) \mid 166 = 2 \cdot 83$, and $2^2 \not\equiv 1 \pmod{167}$, so $83 \mid \text{ord}_{167}(2)$, meaning $\text{ord}_{167}(2) \nmid 75!$. Thus, the answer is $\boxed{167}$.

SR 7: At FenZhu High School, 7th graders have a 60% chance of having a dog and 8th graders have a 40% chance of having a dog, independently of each other. Suppose there is a classroom of 30 7th graders and 10 8th graders. If exactly one person owns a dog, then the probability that a 7th grader owns the dog can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{58}$

Solution: The probability that a 7th grader has the only dog is

$$A = \left(30 \cdot \left(\frac{2}{5}\right)^{29} \cdot \frac{3}{5}\right) \cdot \left(\frac{3}{5}\right)^{10},$$

and the probability for an 8th grader is

$$B = \left(\frac{2}{5}\right)^{30} \cdot \left(10 \cdot \left(\frac{3}{5}\right)^{9} \cdot \frac{2}{5}\right).$$

Then,

$$\frac{A}{B} = \frac{30 \cdot 2^{29} \cdot 3 \cdot 3^{10}}{2^{30} \cdot 10 \cdot 3^9 \cdot 2} = \frac{30 \cdot 3 \cdot 3}{2 \cdot 10 \cdot 2} = \frac{27}{4}.$$

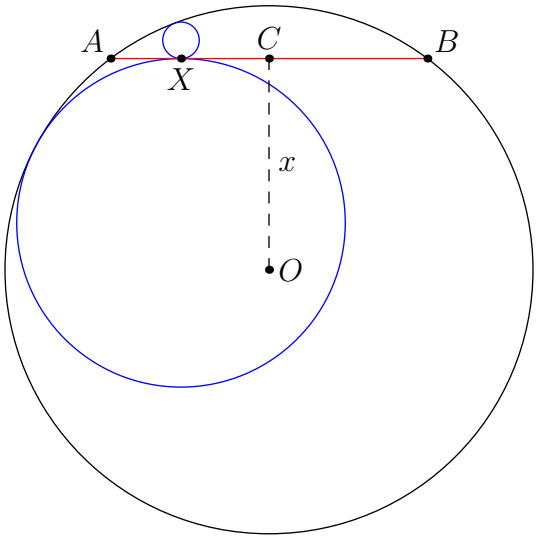
The desired probability is

$$\frac{A}{A+B} = \frac{27}{31} \implies 27 + 31 = \boxed{58}.$$

SR 8: Circle ω has chord AB of length 18. Point X lies on chord AB such that $AX = 4$. Circle ω_1 with radius r_1 and ω_2 with radius r_2 lie on two different sides of AB. Both ω_1 and ω_2 are tangent to AB at X and internally tangent to ω. The sum of the maximum and minimum values of $r_1 r_2$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{1649}$

Solution:



Let O, O_1, O_2 be the centers of $\omega, \omega_1, \omega_2$, respectively. Let r be the radius of ω, let C be the midpoint of AB, and suppose $OC = x$, so $r^2 - x^2 = 81$. Since $AX = 4$ and $AB = 18$, we have $XC = \frac{AB}{2} - AX = 5$. By the Pythagorean Theorem,

$$(O_1X + CO)^2 + (XC)^2 = (OO_1)^2,$$
$$(O_2X - CO)^2 + (XC)^2 = (OO_2)^2,$$

which is equivalent to

$$(r_1 + x)^2 + 25 = (r - r_1)^2,$$
$$(r_2 - x)^2 + 25 = (r - r_2)^2.$$

Solving for r_1 and r_2, we obtain

$$r_1 = \frac{r^2 - x^2 - 25}{2(r + x)},$$
$$r_2 = \frac{r^2 - x^2 - 25}{2(r - x)}.$$

Thus,

$$r_1 r_2 = \frac{((r^2 - x^2) - 5^2)^2}{4(r^2 - x^2)} = \frac{784}{81},$$

which does not depend on ω. So, the maximum and minimum are both $\frac{784}{81}$, giving an answer of

$$\frac{784}{81} + \frac{784}{81} = \frac{1568}{81} \implies 1568 + 81 = \boxed{1649}.$$

SR 9: Find the sum of the maximum and minimum values of $8x^2 + 7xy + 5y^2$ under the constraint that $3x^2 + 5xy + 3y^2 = 88$.

Answer: $\boxed{688}$

Solution: We want the extreme values of k for which there exists a real ratio $a = \frac{x}{y}$ satisfying

$$8a^2 + 7a + 5 = \frac{k}{88}(3a^2 + 5a + 3).$$

Multiplying out and collecting terms gives a quadratic in a whose discriminant simplifies to $11(-k^2 + 688k - 78144)$. For real a, we need $-k^2 + 688k - 78144 \geq 0$, so the extreme values of k are the two roots of $k^2 - 688k + 78144 = 0$. From Vieta's formulas, their sum is $\boxed{688}$.

SR 10: In a circle centered at O with radius 7, we have non-intersecting chords AB and DC. O is outside of quadrilateral $ABCD$ and $AB < CD$. Let X be the intersection of lines AO and CD, and let Y be the intersection of lines BO and CD. Suppose that $XO + YO = 7$. If $YC - DX = 2$ and $XY = 3$, then AB can be expressed as $\frac{a\sqrt{b}}{c}$, where a, b, c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a + b + c$.

Answer: $\boxed{1072}$

Solution:

Let $CY = x + 2$ and $DX = x$. By power of a point, we have

$$(x + 2)(x + 3) = 49 - YO^2 \quad \text{and} \quad (x + 5)x = 49 - XO^2,$$

and subtracting gives $XO^2 - YO^2 = 6$. Since $XO + YO = 7$, dividing gives $XO - YO = \frac{6}{7}$, so $XO = \frac{55}{14}$ and $YO = \frac{43}{14}$. Then, by the Law of Cosines, it follows that

$$OY^2 + OX^2 - 2 \cdot OX \cdot OY \cos \angle YOX = YX^2,$$

which implies that $\cos \angle YOX = \frac{311}{473}$. Then,

$$AB^2 = OA^2 + OB^2 - 2 \cdot OA \cdot OB \cos \angle YOX,$$

which implies that

$$AB = \frac{126\sqrt{473}}{473} \implies 126 + 473 + 473 = \boxed{1072}.$$